

Explaining Race and Gender Wage Gaps: An OLS Analysis of Human Capital, Background, and Labor Market Factors Using NLSY79 Data

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Abstract

Background: This paper examines race and gender wage disparities using the National Longitudinal Survey of Youth 1979 panel, comparing white men with white women, Hispanic men and women, and Black men and women. **Methods:** The analysis uses Ordinary Least Squares regressions of log hourly wages to evaluate whether baseline wage gaps are explained by human capital variables, background factors, and current labor-market or life-circumstance variables. **Results:** Baseline models show lower wages for all non-white-male groups. Adding schooling, age, tenure, parental education, poverty status in 1979, AFQT percentile, marital status, and industry controls explains a substantial share of the initial gaps. The adjusted R^2 rises from 0.050 in the baseline log-wage model to 0.329 in the final specification, while the final statistically significant remaining race-gender coefficient is concentrated among white women. **Conclusions:** The results suggest that observed wage disparities are connected not only to measurable human capital differences, but also to background inequality and labor-market structure. The paper concludes with policy recommendations focused on early education access, childhood poverty, college affordability, pay transparency, anti-discrimination enforcement, and future research on career interruptions and employer bias.

Keywords: wage gaps; race; gender; human capital; OLS; NLSY79; AFQT; labor-market inequality

JEL Classification: J31; J16; J15; C21; I24

Article Type: Empirical research paper

Data Source: National Longitudinal Survey of Youth 1979 panel, using the sample and variable restrictions described in the manuscript.

The full manuscript begins on the following page. The original text, tables, coefficients, sample sizes, and numerical results are preserved.

Explaining Race and Gender Wage Gaps: An OLS Analysis of Human Capital, Background, and Labor Market Factors Using NLSY79 Data

This project explores wage disparities between white men and five other racial/gender groups; white women, hispanic men/women, black men/women, using data from the National Longitudinal Survey of Youth (NLSY) 1979 panel. It estimates the extent of these wage gaps and examines whether they can be explained by differences in human capital and background characteristics. To establish a baseline, average hourly wages by race and gender are reported below.

Regression Results Wage

	(1)
	WAGE
1.RACE#0.MALE1	0.000
	(.)
1.RACE#1.MALE1	6.850*
	(3.403)
2.RACE#0.MALE1	-1.202
	(3.134)
2.RACE#1.MALE1	-0.117
	(3.145)
3.RACE#0.MALE1	-0.173
	(2.873)
3.RACE#1.MALE1	8.153**
	(2.842)
_cons	16.422***
	(2.514)
adj. R^2	0.015
F	6.439
rmse	29.638
N	1757

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Three hypotheses guide this investigation. Human Capital Theory (HKT) proposes that wages reflect accumulated education, training and work experience, so unexplained gaps should shrink once variables like years of schooling, age (proxy for experience), and tenure are included. Expanding on this, background variables (Cordero-Guzman 2001) highlight how parental education, stable family structure, and regional schooling conditions influence development and educational attainment. This challenges interpretations of wage gaps as merit-based, emphasizing how background characteristics shape opportunities before entry into the labor market. Thirdly, differences in current life circumstances, including marital status and industry of employment reflect structural labor differences or gender traditions that influence wages independently of HKT and background variables. The table above presents statistics for each race-gender group, showing disparities in both wages and background characteristics. White men, on average, earn more than every other group in the sample, with particularly large gaps separating them from Black and Hispanic women. The baseline regression confirms this pattern that all non-white-male groups earn significantly less. To analyze the cause of this gap, I excluded white men (NBNH_Male) from the model as the reference group in regressions.

To test these hypotheses, the Ordinary Least Squares method of regression analysis is used, which minimizes the sum of squared residuals; the distances between actual and predicted values, ensuring that the fitted regression line best explains the variation in average hourly wages amongst individuals in the NLSY79. The population is respondents to the NLSY79 dataset, restricted to those with valid wage observations. Initial explanatory variables for HKT are race-gender dummies, years of school, their interactions, age, age², and tenure in years. The response variable, log(wage), allows coefficients to be interpreted as percent changes. The primary parameter of interest are the coefficients of race-gender groups, representing the

unexplained wage gap after accounting for measurable differences amongst groups. Accounting for Guzman's emphasis on background components, variables measuring parental education, poverty status in 1979, and AFQTPctile are included. Parents' level of education may influence children's schooling through both income and human capital absorption. Poverty status in 1979 captures both family income and school district quality. Schools in impoverished areas have less resources, with students in poverty having limited access to private education or tutors.

AFQTPctile serves as a proxy on students' learning environment through test scores. Marital status male/female are included to account for marriage benefits hypothesized with men due to gender norms and expectations around household labor and caregiving. Industry categories are included to reduce omitted variable bias, which occurs when an omitted variable is relevant, and correlates with included regressors. Together, these variables allow for evaluating if wage differentials are explained by measurable differences in productivity and background, or if unexplained gaps point toward persistent structural inequality.

Summary Statistics for White Men

	Obs	Mean	Standard Deviation	minimum	maximum
WAGE	354	24.29	(28.59)	0.44	352.56
S	354	13.77	(2.59)	7.00	20.00
AGE	354	40.24	(7.71)	26.00	57.00
TENURE1	354	359.08	(352.19)	1.00	1,864.00
AFQTPctile	354	55.35	(28.04)	0.00	100.00
S_M	354	11.64	(2.68)	0.00	20.00
S_F	354	11.78	(3.63)	0.00	20.00
POVSTATUS_1979	354	0.07	(0.26)	0.00	1.00
MM	354	0.64	(0.48)	0.00	1.00
MF	354	0.00	(0.00)	0.00	0.00
CONST_MANU	354	0.43	(0.50)	0.00	1.00
SALES	354	0.15	(0.36)	0.00	1.00
SERVICES	354	0.32	(0.47)	0.00	1.00
N	354				

Summary Statistics for White Women

	Obs	Mean	Standard Deviation	minimum	maximum
WAGE	314	16.31	(14.18)	0.29	115.38
S	314	13.95	(2.31)	9.00	20.00
AGE	314	40.94	(7.87)	26.00	58.00
TENURE1	314	336.60	(334.73)	1.00	1,641.00
AFQTPctile	314	55.29	(26.87)	0.25	99.82
S_M	314	11.91	(2.66)	0.00	18.00
S_F	314	12.41	(3.25)	0.00	20.00
POVSTATUS_1979	314	0.10	(0.30)	0.00	1.00
MM	314	0.00	(0.00)	0.00	0.00
MF	314	0.66	(0.47)	0.00	1.00
CONST_MANU	314	0.19	(0.39)	0.00	1.00
SALES	314	0.15	(0.36)	0.00	1.00
SERVICES	314	0.58	(0.49)	0.00	1.00
<i>N</i>	314				

Summary Statistics for Hispanic Men

	Obs	Mean	Standard Deviation	minimum	maximum
WAGE	103	16.53	(10.77)	3.75	62.50
S	103	12.29	(2.41)	4.00	20.00
AGE	103	40.12	(7.71)	26.00	57.00
TENURE1	103	337.92	(332.87)	1.00	1,185.00
AFQTPctile	103	33.24	(25.43)	0.00	90.91
S_M	103	7.95	(4.11)	0.00	17.00
S_F	103	7.65	(4.37)	0.00	16.00
POVSTATUS_1979	103	0.33	(0.47)	0.00	1.00
MM	103	0.58	(0.50)	0.00	1.00
MF	103	0.00	(0.00)	0.00	0.00
CONST_MANU	103	0.50	(0.50)	0.00	1.00
SALES	103	0.12	(0.32)	0.00	1.00
SERVICES	103	0.23	(0.42)	0.00	1.00
<i>N</i>	103				

Summary Statistics for Hispanic Women

	Obs	Mean	Standard Deviation	minimum	maximum
WAGE	100	17.52	(21.13)	0.27	175.00
S	100	13.08	(2.65)	3.00	20.00
AGE	100	40.36	(7.43)	26.00	57.00
TENURE1	100	285.60	(303.48)	2.00	1,418.00
AFQTPctile	100	32.91	(21.35)	0.26	82.48
S_M	100	7.82	(3.99)	0.00	16.00
S_F	100	8.33	(4.51)	0.00	19.00
POVSTATUS_1979	100	0.30	(0.46)	0.00	1.00

MM	100	0.00	(0.00)	0.00	0.00
MF	100	0.49	(0.50)	0.00	1.00
CONST_MANU	100	0.14	(0.35)	0.00	1.00
SALES	100	0.18	(0.39)	0.00	1.00
SERVICES	100	0.58	(0.50)	0.00	1.00
<i>N</i>	100				

Summary Statistics for Black Men

	Obs	Mean	Standard Deviation	minimum	maximum
WAGE	132	16.77	(27.05)	0.03	300.00
S	132	13.04	(2.08)	8.00	20.00
AGE	132	40.29	(7.51)	27.00	57.00
TENURE1	132	359.85	(362.72)	4.00	1,580.00
AFQTPctile	132	28.82	(24.49)	0.00	100.00
S_M	132	11.27	(2.58)	5.00	19.00
S_F	132	10.36	(3.38)	0.00	18.00
POVSTATUS_1979	132	0.27	(0.44)	0.00	1.00
MM	132	0.37	(0.48)	0.00	1.00
MF	132	0.00	(0.00)	0.00	0.00
CONST_MANU	132	0.45	(0.50)	0.00	1.00
SALES	132	0.12	(0.33)	0.00	1.00
SERVICES	132	0.33	(0.47)	0.00	1.00
<i>N</i>	132				

Summary Statistics for Black Female

	Obs	Mean	Standard Deviation	minimum	maximum
WAGE	147	17.16	(37.22)	1.92	437.50
S	147	13.54	(2.07)	7.00	20.00
AGE	147	41.74	(7.61)	28.00	57.00
TENURE1	147	335.95	(327.11)	1.00	1,525.00
AFQTPctile	147	28.70	(21.40)	0.24	88.02
S_M	147	11.26	(2.72)	3.00	18.00
S_F	147	10.31	(3.90)	0.00	20.00
POVSTATUS_1979	147	0.31	(0.47)	0.00	1.00
MM	147	0.00	(0.00)	0.00	0.00
MF	147	0.37	(0.48)	0.00	1.00
CONST_MANU	147	0.19	(0.39)	0.00	1.00
SALES	147	0.09	(0.28)	0.00	1.00
SERVICES	147	0.61	(0.49)	0.00	1.00
<i>N</i>	147				

The summary of statistics tables show substantial wage disparities by group, with white men earning the highest average hourly wage (\$24.29), and white women having the lowest (\$16.31). Black women display a potential outlier, having the highest maximum hourly wage at \$437.50, but a mean wage of \$17.16 (only third largest), supporting our logarithmic transformation of wages to handle this outlier. Hispanic men have a year less of education compared to other groups, with a mean of 12.29 years of education and minimum of 4 years. Hispanic men and women also have the lowest parental education (7-8 years), demonstrating an intergenerational disadvantage. Poverty rates are lowest amongst the White respondents, with averages of 10% for females and 7% for males. They also performed better on the AFQT, staying at a mean of 55th percentile for both groups, whilst the others fell around 20-30th percentiles. Another potential outlier could be seen in the Black Male AFQT scores, as the maximum score is 100, yet they have the lowest mean at 28.57.

OLS regression estimates the effect of human capital and background characteristics on the log of hourly wages, helping address OVB. These regressions rely on the Classical Linear Model assumptions. The first five are the Gauss Markov assumptions, used to determine unbiased OLS estimators, where the expected value of the estimated coefficients equals the true population parameter, as well as achieve the lowest variance amongst all linear unbiased estimators. An additional assumption is included as well to determine normality. Of particular importance is the zero conditional mean assumption, requiring the error term to have an expected value of zero given the explanatory variables. If violated, the estimated coefficient will be biased. Although the model includes variables for education, tenure, age, family background and income and neighborhood region to reduce bias, unobserved biases may still be correlated with wages and the regressors. No perfect collinearity is also critical, stating no explanatory variable is

constant or a perfectly linear function of another, met by omitting a base group from categorical variables such as white men from race-gender groups and public administration for industry categories. While the normality of the error term is not assumed, the Central Limit Theorem guarantees that in large samples, the sampling distribution of the OLS estimates resembles a normal distribution. This allows for hypothesis testing using the t-distribution; $t_j = \beta_j - \beta_j / \text{se}(\beta_j) \sim t_{n-k-1}$, allows us to conduct hypothesis testing of the null: $H_0 = \beta_j = 0$, and an alternative hypothesis: $H_A = \beta_j \neq 0$. In this model, we use two-sided tests and a 5% significance level. Large t-statistics in comparison to the critical value of 1.96 leads to a rejection of null and statistical significance, showing that the variable does have an effect on $\log(\text{wage})$. A stronger way to analyze the strength of a variable against the null is its p-value, which represents the probability of observing a t-statistic as extreme as the one achieved if the null was true. A p-value less than 0.05 indicates strong evidence against the null hypothesis, suggesting the coefficient's statistical significance. Law of large numbers further supports the reliability of estimates by ensuring convergence of sample means to population means as sample size increases.

Regression Results lwage

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	lwage	lwage	lwage	lwage	lwage	lwage	lwage	lwage	lwage	lwage	lwage	lwage
NBNH_Female	-0.350*** (0.049)	-0.372*** (0.047)	-0.074 (0.262)	-0.168 (0.245)	-0.173 (0.245)	-0.198 (0.240)	-0.367*** (0.043)	-0.362*** (0.049)	-0.332*** (0.048)	-0.147* (0.058)	-0.163* (0.067)	-0.134* (0.067)
HISP_Male	-0.230*** (0.068)	-0.080 (0.065)	0.569 (0.320)	0.329 (0.300)	0.325 (0.300)	0.299 (0.293)	-0.123 (0.069)	-0.097 (0.074)	0.012 (0.067)	0.023 (0.066)	0.022 (0.066)	0.017 (0.066)
HISP_Female	-0.396*** (0.073)	-0.306*** (0.069)	0.508 (0.339)	0.184 (0.318)	0.179 (0.318)	0.189 (0.310)	-0.288*** (0.069)	-0.271*** (0.075)	-0.177* (0.070)	0.002 (0.076)	-0.012 (0.081)	0.023 (0.082)
BLCK_Male	-0.425*** (0.059)	-0.341*** (0.056)	0.101 (0.334)	-0.140 (0.312)	-0.147 (0.313)	-0.173 (0.305)	-0.345*** (0.057)	-0.349*** (0.065)	-0.172** (0.062)	-0.101 (0.062)	-0.102 (0.062)	-0.104 (0.062)
BLCK_Female	-0.454*** (0.059)	-0.416*** (0.056)	-0.347 (0.325)	-0.715* (0.305)	-0.718* (0.305)	-0.710* (0.301)	-0.427*** (0.056)	-0.379*** (0.063)	-0.277*** (0.061)	-0.094 (0.068)	-0.104 (0.071)	-0.079 (0.071)
S		0.100*** (0.007)	0.121*** (0.012)	0.103** (0.011)	0.103** (0.011)	0.098** (0.011)	0.084*** (0.007)	0.079*** (0.008)	0.053*** (0.009)	0.050** (0.009)	0.050** (0.009)	0.053** (0.009)

S_NBNHFemale			-0.022	-0.016	-0.015	-0.012						
			(0.019)	(0.018)	(0.018)	(0.017)						
S_HISPMale			-0.051*	-0.033	-0.032	-0.033						
			(0.025)	(0.024)	(0.024)	(0.023)						
S_HISPFemale			-0.062*	-0.038	-0.038	-0.037						
			(0.026)	(0.024)	(0.024)	(0.023)						
S_BLACKMale			-0.033	-0.015	-0.014	-0.012						
			(0.025)	(0.024)	(0.024)	(0.023)						
S_BLACKFemale			-0.005	0.019	0.019	0.019						
			(0.024)	(0.022)	(0.023)	(0.022)						
AGE				0.033** *	0.050*	0.054*	0.074**	0.074**	0.061*	0.051*	0.051*	0.050*
				(0.002)	(0.022)	(0.022)	(0.023)	(0.026)	(0.024)	(0.024)	(0.024)	(0.024)
AGEsq					-0.000	-0.000	-0.001*	-0.001	-0.000	-0.000	-0.000	-0.000
					(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
TENURE_YR						0.021** *	0.021***	0.023***	0.022***	0.020** *	0.020** *	0.019** *
						(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
S_M							-0.001	-0.002				
							(0.007)	(0.008)				
S_F							0.001	0.002				
							(0.006)	(0.006)				
POVSTATUS_1979								-0.095	-0.095*	-0.088	-0.086	-0.091*
								(0.052)	(0.046)	(0.045)	(0.046)	(0.045)
AFQTPctile									0.005***	0.005** *	0.005** *	0.006** *
									(0.001)	(0.001)	(0.001)	(0.001)
MM										0.287** *	0.287** *	0.278** *
										(0.050)	(0.050)	(0.050)
MF											0.026	0.026
											(0.052)	(0.052)
CONST_MANU												0.023
												(0.064)
SALES												-0.188*
												(0.073)
SERVICES												-0.106
												(0.062)
_cons	2.877***	1.507***	1.229***	0.137	-0.188	-0.235	-0.402	-0.373	-0.099	-0.017	-0.017	0.006
	(0.034)	(0.102)	(0.167)	(0.171)	(0.462)	(0.454)	(0.460)	(0.516)	(0.488)	(0.482)	(0.482)	(0.489)
adj. R ²	0.050	0.148	0.150	0.257	0.256	0.284	0.294	0.294	0.307	0.322	0.322	0.329
F	19.498	51.815	29.160	51.513	47.582	50.106	55.704	42.114	55.945	55.174	50.920	42.932
rmse	0.759	0.719	0.718	0.672	0.672	0.653	0.627	0.634	0.645	0.638	0.638	0.635
N	1757	1757	1757	1757	1757	1734	1446	1187	1368	1368	1368	1368

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

I first regressed the five race/gender categorical groups, measuring the $\log(\text{wages})$ on and the five race-gender groups. All group coefficients are statistically significant with p-values less than 0.05. Implying, on average, White women earn 35% less, Hispanic men 23% less, Hispanic women 39.6% less, Black men 42.5% less, and Black women 45.5% less than White men. The R^2 value of 0.05 indicates that these race-gender groups explain only 5% of variation in hourly wages.

The second regression adds the first human capital variable, S (years of schooling). Under HKT, more education increases wages. Including schooling reduces all race-gender coefficients and removes statistical significance from Hispanic men, suggesting part of the initial wage gaps was from differences in educational attainment. Next, interaction terms between schooling and race-gender groups (estimating group-specific return of schooling) are included, causing substantial shifts in race-gender dummy coefficients, and further losses in statistical significance. These shifts in coefficients signal OVB in the initial model, where part of the observed wage gap was due to educational differences, where the bias is downward. However, because of the interaction terms, these coefficients no longer measure wage gaps directly. Instead, coefficients on the race-gender groups shift the baseline intercept, the constant in the regression (intercept of white men), while the interactions adjust the slope on schooling for each group. The coefficient on schooling, 0.121, represented the estimated return to an additional year of schooling for white men. So taking a variable like Black Men, the dummy and its interaction with schooling imply a new intercept and slope for Black men. Their predicted wage function has an intercept of 1.330 and a slope of 0.088. Although the slope adjustment is statistically insignificant, a joint F-test confirms that the intercept and slope together are significantly

different from white men's, suggesting differences in wage determination for this group. Most interactive variables are negative and statistically insignificant (aside from Hispanic men/female), suggesting a lower return of schooling for each racial-gender group against white men. Due to the statistical insignificance of most race-gender groups, a joint F test was run on each race-gender group individually and their interactive variable counterparts. The null hypothesis that the race-gender and their interactive variables are jointly equal to zero were all rejected at the 5% level, with F-statistics larger than 10, and p-values lower than 0.05, suggesting their joint significance in this regression and supporting their inclusion in the model. However, The F-test for Hispanic men failed to reject the null, with an F-statistic of 2.57 and p-value of 0.077, meaning we cannot conclude that their wage function jointly differs from that of white men. But since the interaction term is significant, the slope of their wage may differ. In this case, the unexplained wage gap must be calculated at the sample mean of schooling of white men rather than from individual coefficients, where Hispanic men earn 13.2% less than white men on average. This suggests that the wage gap seen in the first model for Hispanic men may be explained primarily by their lower average years of schooling, consistent with their summary statistics showing the lowest average educational attainment among all groups. The R^2 rises to 0.150, so all group variables, schooling, and interactive variables explains 15% of variation in hourly wages.

The next regression adds age, another HKT variable used as a proxy for experience. It is statistically significant and the estimated coefficient indicates that on average, each additional year of age is associated with a 3.3% increase in hourly wages. Controlling for age revealed a larger and statistically significant wage penalty for Black women, decreasing its coefficient to -0.715, suggesting that differences in experience obscured part of their wage gap. Their return of

schooling coefficient increased 0.019, indicating that that education becomes more strongly associated with wages for Black women once age is accounted for. To capture the diminishing marginal effects of age on $\log(\text{wage})$, since I hypothesize wages rise with experience but level off later in life, I added age's quadratic transformation. The estimated coefficient on age changed to 0.05, remaining statistically significant, and its quadratic having a negative coefficient, suggesting a concave relationship so that hourly wages rise with age at a decreasing rate. At the sample mean age of 40, the estimated slope was 0.028, indicating that at 40 years, a one year increase in age is associated with a 2.8% increase in hourly wages.

The last human capital variable included in the regression is tenure in years. Tenure's estimated coefficient is statistically significant, indicating that for every additional year of tenure, hourly wages increase by 2.1%. After including age and tenure, only the coefficients on Black women, age, schooling and tenure remained statistically significant. Joint F-tests on each race-gender dummy and their interaction term showed the same outputs as the first regression; all but Hispanic men variables are jointly statistically significant.

The next variables included in the regression are background variables to measure how circumstances growing up affect human capital. The interactive variables for return of school for each race-gender group are excluded from the following regressions to isolate these effects and avoid multicollinearity. First, S_M (Mothers education) and S_F (Fathers education) are added as proxies for household educational environments. Both have positive but statistically insignificant coefficients suggesting each additional year of parental education is associated with a 5% increase in hourly wages. All other variables, excluding Hispanic men and $AGEsq$, remain significant, likely due to the removal of previous interactive variables and inclusion of S_M/S_F .

An F-test was run on S_M and S_F, showing joint insignificance between the two variables, supporting their exclusion in the model after checking their use in the regression with poverty.

Poverty status is added to account for household economic disadvantages during adolescence, as those in impoverished areas likely had less access to quality education and resources. The variable is binary, equal to 1 if the respondent lived in poverty in 1979. The estimated coefficient on poverty status is negative and statistically insignificant, implying that individuals who grew up in poverty earn 9.5% less in hourly wages on average. Its statistical insignificance is likely due to multicollinearity between parental education and poverty, which is proved when parental education is removed. This leads to the decision to remove parental education from further regressions. Poverty inclusion increases most of the dummy variables except Black men's, suggesting part of their observed wage gaps in earlier regressions were explained by differences in childhood poverty.

The last background variable is AFQT percentile score, a proxy for accumulated human capital and early access to education resources, as argued by Cordero-Guzmán. The estimated coefficient on AFQT percentile is statistically significant, and estimates a 0.5% increase in hourly wage for each 1 increase in percentile score of the AFQT. The estimated coefficients on all race-gender dummies had substantial increases, except for White women, indicating differences in AFQT, representing accumulated human capital due to available learning resources, are important in explaining the wage disparities between white men and the race-gender groups. However, given that many of the wage disparities are explained by the previous background variables as well, it only reinforces that poverty and differences in background are causing this effect. The estimated coefficient on schooling fell as well, 0.79 to

0.53, indicating upward bias through OVB. This suggests that part of schoolings return reflected reflected differences in learning environments captured by AFQT.

The final variables included in the regression are current variables, testing factors like marital status and employment in key occupational sectors. The first variable included in the regression is the interactive variable for married men, in comparison to unmarried men and married/unmarried women. The coefficient on married men is statistically significant, indicating marriage for males on average is associated with 28.7% higher hourly wages compared to single men and married/unmarried women. The estimated coefficients on the race-gender dummies all increased significantly, aside from Hispanic men and Black men, indicating the effect of marriage on wages is far less pronounced among married Hispanic and Black men. The interactive variable, married females, is added to measure the effect of marriage on the wages for women. The coefficient is statistically insignificant, and slightly decreased the coefficients of all 6 dummy variables, but still indicating the wage gap between married men and women is larger than between single men and women. This insignificance suggests multicollinearity between male and female marriage and offsetting factors like household burdens and employment discrimination. While MM is individually significant, we conducted an F-test on both MM and MF to assess their joint significance, given MF's insignificance. The results suggest that together, the two variables significantly contribute to the model.

The next current variable looks at how wage is related to current occupation. Public administration is omitted to serve as a baseline category, so the estimated coefficients on the other variables represent wage differentials relative to this industry. Among the sectors included, only sales is statistically significant, with an estimated coefficient of -0.188, suggesting that respondents working in Sales earn 18.8% less per hour than those employed in public

administration on average. The coefficients on both construction/manufacturing and services are both negative as well, and their statistical insignificance indicates we cannot conclude there is a wage gap relative to public administration. The addition of these occupation variables did not substantially change the coefficients or standard errors of the race-gender dummy variables, supporting their addition to control for possible occupational sorting by race/gender. Black and White females remain the only coefficients of the dummy variables to remain statistically significant. Poverty gains statistical significance in this regression, with an estimated coefficient indicating a 9.1% decrease in hourly wages through impoverished adolescence. Thus including occupation variables further sharpens the effect estimates of background and demographic factors. The R^2 for this final regression increased to 0.329, so that all variables included in this regression explain 32.9% variation in hourly wages, a significant increase from the initial R^2 of 0.05.

By the end of the regressions, most of the wage gaps between white men and other racial/gender groups were explained by observable factors in the NYSL dataset, with the only statistically significant estimated coefficient being on White women. Through these regressions it is clear human capital variables are a product of the background variables seen throughout the regressions. The poverty and AFQT percentile scores were significant predictors of wage, and more importantly help explain how unequal early life environments shape long term earnings potential. These factors moderate the ability to accumulate human capital, contributing racial/ethnic disparities in wages. Marital status was also a strong predictor of wage, as both married men/women have increased drive to make money to support their respective families. However the gap between the two variables suggest additional factors, such as discrimination against married women or traditional marital norms. Other limitations on this study involve the

absence of direct measures of work interruptions and employer bias. My policy recommendations are to focus on expanding early education access, addressing childhood poverty with greater government aid, increasing college affordability by increasing FAFSA funding, and enforcing pay transparency with anti discriminatory laws to better focus on employment inequalities and workplace disparities. Finally, because some group differences remained even after controls, future research should include measures of career interruptions and employer bias to better understand persistent gaps.

References:

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